

3.5 'How Science Works'

How Science Works is an underpinning set of concepts and is the means whereby students come to understand how scientists investigate scientific phenomena in their attempts to explain the world about us. Moreover, How Science Works recognises the contribution scientists have made to their own disciplines and to the wider world.

Further, it recognises that scientists may be influenced by their own beliefs and that these can affect the way in which they approach their work. Also, it acknowledges that scientists can and must contribute to debates about the uses to which their work is put and how their work influences decision-making in society.

In general terms, it can be used to promote students' skills in solving scientific problems by developing an understanding of

- The concepts, principles and theories that form the subject content
- The procedures associated with the valid testing of ideas and, in particular, the collection, interpretation and validation of evidence
- The role of the scientific community in validating evidence and also in resolving conflicting evidence

More details of these principles and procedures are given in (A) to (I) below and these are taken from paragraph 3.6 of GCE AS and A level subject criteria for science (September 2006) which sets out the

requirements for the skills, knowledge and understanding of How Science Works.

When students are proficient in dealing with the matters given in (A) to (I) we may say that they have achieved a level of scientific literacy: they have learned the 'language' of science. Students are then, and only then, able to engage with the place and contribution of science in the wider world. (See (J) to (L) below, again taken from paragraph 3.6 of the science criteria.) In particular, students will begin to recognise:

- The contribution that scientists as scientists can make to decision-making and the formulation of policy
- The need for regulation of scientific enquiry and how this can be achieved
- How scientists can contribute legitimately in debates about those claims which are made in the name of science.

Clearly, an examination specification, covering a given body of scientific content and designed for students working at a particular level, cannot cover to the same depth and comprehensiveness all the issues dealt with under (A) to (L) below as these describe in general terms scientific activity which permeates all scientific disciplines. This specification will concentrate on those aspects which, it is deemed, are most appropriate at this level and taking into account the content to be studied. In doing so, it

bears in mind considerations of what is reasonable, manageable and practicable. So, under each of (A) to (L) is indicated the general approach by which the specification will attempt to deal with the individual concepts and associated activities. Also, brief, illustrative examples of areas of the specification where the particular idea(s) may be assessed are included. These examples are not exhaustive and, therefore, do not prescribe every area where How Science Works will be assessed.

A Use theories, models and ideas to develop and modify scientific explanations.

Scientists use theories and models to attempt to explain observations. These theories or models can form the basis for scientific experimental work.

Scientific progress is made when validated evidence is found that supports a new theory or model.

Candidates will be expected to:

- propose hypotheses and design appropriate studies based on their understanding of psychological theory and models
- explain results of studies in terms of psychological theory and concepts
- compare different theories in terms of their ability to explain psychological processes
- apply psychological knowledge to unfamiliar situations.

Examples in this specification include the following:

- *Students might use their knowledge of the multi-store model of memory to design a study to test the effects of rehearsal (§3.2.3).*
- *Candidates might apply their knowledge of social cognition to explain aspects of behaviour presented in a scenario question about a person attending an interview (§3.2.2).*

B Use knowledge and understanding to pose scientific questions, define scientific problems, present scientific arguments and scientific ideas.

Scientists use their knowledge and understanding when observing objects and events, in defining a scientific problem and when questioning the explanations of themselves or of other scientists.

Scientific progress is made when scientists contribute to the development of new ideas, materials and theories.

Candidates will be expected to use knowledge of psychological theory and research to:

- propose psychological hypotheses

- identify appropriate issues for psychological investigations
- explain psychological processes and concepts
- present informed psychological arguments.

Examples in this specification include the following:

- *Students might use their knowledge and understanding of cognitive psychology to propose a series of experimental hypotheses about perception (§3.2.4).*
- *Candidates might review findings of research into the effects of expectation on perception to present an argument in favour of Gregory's theory of perception (§3.2.4).*

C Use appropriate methodology, including ICT, to answer scientific questions and solve scientific problems.

Observations ultimately lead to explanations in the form of hypotheses. In turn, these hypotheses lead to predictions that can be tested experimentally. Observations are one of the key links between the 'real world' and the abstract ideas of science.

Once an experimental method has been validated, it becomes a protocol that is used by other scientists.

ICT can be used to speed up, collect, record and analyse experimental data.

Candidates will be expected to:

- choose methods appropriate for the investigation of psychological problems
- use ICT to prepare materials for an investigation
- use ICT to analyse and present data from psychological investigations.

Examples in this specification include the following:

- *Students might choose to investigate differences in gender-related behaviour, preparing a suitable questionnaire using a word processing package (§3.1.3).*
- *Candidates might be required to choose an appropriate inferential statistical test to analyse sets of data from a research study (§3.4.4).*

D Carry out experimental and investigative activities, including appropriate risk management in a range of contexts.

Scientists perform a range of experimental skills that include manual and data skills (tabulation, graphical skills, etc).

Scientists should select and use equipment that is appropriate when making accurate measurements and should record these measurements methodically.

Scientists carry out experimental work in such a way as to minimise the risk to themselves and others.

Candidates will be expected to:

- take part in class investigations using the experimental and other methods to investigate psychological theories, models and ideas
- consider possible physical and psychological risk to all group members and adjust activities accordingly.

Examples in this specification include the following:

- *Students might conduct a class experiment on classical conditioning of the eye-blink response using a puff of air as the unconditioned stimulus and noise as the neutral-conditioned stimulus (§3.1.1).*
- *Candidates might discuss the physical and psychological risks associated with carrying out a lifestyle survey using an opportunity sample in a town centre (§3.1.4).*

E Analyse and interpret data to provide evidence, recognising correlations and causal relationships.

Scientists look for patterns and trends in data as a first step in providing explanations of phenomena. The degree of uncertainty in any data will affect whether alternative explanations can be given for the data.

Anomalous data are those measurements that fall outside the normal, or expected, range of measured values. Decisions on how to treat anomalous data should be made only after examination of the event.

In searching for causal links between factors, scientists propose predictive theoretical models that can be tested experimentally. When experimental data confirm predictions from these theoretical models, scientists become confident that a causal relationship exists.

Candidates will be expected to:

- analyse and interpret raw and summary data from psychological investigations in class
- explain and interpret summary data from psychological investigations in examinations
- interpret and draw conclusions from findings of their own and others' investigations
- understand the difference between correlation and cause and effect relationships.

Examples in this specification include the following:

- *Students might collect two sets of data in class (digit span and androgyny score), correlate the two sets of data, then discuss the reasons why correlation cannot be used to infer cause and effect (§3.1.4).*

- *Candidates might be given a table of summary data from an investigation into social facilitation and asked to draw conclusions in relation to the aim of the study described, or in relation to psychological theory (§3.2.1).*

F Evaluate methodology, evidence and data, and resolve conflicting evidence.

The validity of new evidence, and the robustness of conclusions that stem from it, is constantly questioned by scientists.

Experimental methods must be designed adequately to test predictions.

Solutions to scientific problems are often developed when different research teams produce conflicting evidence. Such evidence is a stimulus for further scientific investigation, which involves refinements of experimental technique or development of new hypotheses.

Candidates will be expected to:

- evaluate psychological methods in terms of issues such as reliability, validity and ethics
- evaluate evidence on the basis of the method used to gather the evidence
- explain anomalous scores in a set of data
- explain inconsistencies in data
- evaluate the use of descriptive and inferential statistics.

Examples in this specification include the following:

- *Students might be given sets of data in class and asked to identify and explain any anomalous scores (§3.4.5).*
- *Candidates might discuss the use of different methods to investigate localisation of function in the brain (§3.1.2).*

G Appreciate the tentative nature of scientific knowledge

Scientific explanations are those that are based on experimental evidence, which is supported by the scientific community.

Scientific knowledge changes when new evidence provides a better explanation of scientific observations.

Candidates will be able to understand that scientific knowledge is founded on experimental evidence and that such evidence must be shown to be reliable and reproducible. If such evidence does not support a theory the theory must be modified or replaced with a different theory. Just as previous scientific theories have been proved inadequate or incorrect, our present theories may also be flawed.

Candidates will be expected to:

- understand the features and principles of the scientific approach in psychology including a paradigm, the role of theory, hypothesis testing, replication and generalisation.

Examples in this specification include the following:

- *Students might carry out their own research into levels of processing, replicate the study two weeks later using the same sample, find that the results are not exactly the same and explore reasons for this (§3.2.3).*
- *Candidates might discuss the behaviourist approach in psychology, considering how the popularity of behaviourism waned leading to the development of the cognitive approach (§3.4.1).*

H Communicate information and ideas in appropriate ways using appropriate terminology.

By sharing the findings of their research, scientists provide the scientific community with opportunities to replicate and further test their work, thus either confirming new explanations or refuting them.

Scientific terminology avoids confusion amongst the scientific community, enabling better understanding and testing of scientific explanations.

Candidates will be expected to:

- use the appropriate psychological terminology to express psychological ideas, describe psychological concepts, interpret and explain psychological findings and describe and discuss psychological theories.

Examples in this specification include the following:

- *Students might design a practical in small groups, then present their designs to the class as a whole, using appropriate psychological terminology (§3.1.4 and §3.4.5).*
- *Candidates should make appropriate reference to Piagetian concepts in a discussion of Piaget's theory of cognitive development (§3.3.2).*

I Consider applications and implications of science and appreciate their associated benefits and risks.

Scientific advances have greatly improved the quality of life for the majority of people. Developments in technology, medicine and materials continue to further these improvements at an increasing rate.

Scientists can predict and report on some of the beneficial applications of their experimental findings.

Scientists evaluate, and report on, the risks associated with the techniques they develop and applications of their findings.

Candidates will be expected to:

- understand and appreciate applications of psychology
- understand and appreciate the implications of psychological findings
- evaluate the contribution of psychological research in terms of benefits and risks.

Examples in this specification include the following:

- *Students might carry out a classroom investigation of eyewitness testimony using two conditions, to compare the effectiveness of the cognitive interview and the standard interview (Geiselman 1985) (§3.3.4).*
- *Candidates might discuss various psychological treatments for psychological disorders, with reference to the benefits and risks to the patient and to society as a whole (§3.3.5).*

J Consider ethical issues in the treatment of humans, other organisms and the environment.

Scientific research is funded by society, either through public funding or through private companies that obtain their income from commercial activities. Scientists have a duty to consider ethical issues associated with their findings.

Individual scientists have ethical codes that are often based on humanistic, moral and religious beliefs.

Scientists are self-regulating and contribute to decision making about what investigations and methodologies should be permitted.

Candidates will be able to appreciate how science and society interact. They should examine how science has provided solutions to problems but that the solutions require society to form judgements as to whether the solution is acceptable in view of moral issues that result. Issues such as the effects on the planet, and the economic and physical well being of the living things on it should be considered.

Candidates will be expected to:

- understand and appreciate the British Psychological Society code of ethics
- apply the code of ethics in their class investigations
- discuss other psychological investigations in relation to the code of ethics.

Examples in this specification include the following:

- *Candidates might design an attitude scale (Likert type) to investigate attitudes to global warming (§3.2.2).*
- *Candidates might discuss ethical issues related to the use of investigations into the genetic basis of behaviour (§3.1.2).*

K Appreciate the role of the scientific community in validating new knowledge and ensuring integrity.

The findings of scientists are subject to peer review before being accepted for publication in a reputable scientific journal.

The interests of the organisations that fund scientific research can influence the direction of research. In some cases the validity of those claims may also be influenced.

Candidates will understand that scientists need a common set of values and responsibilities. They should know that scientists undertake a peer-review of the work of others. They should know that scientists work with a common aim to progress scientific knowledge and understanding in a valid way and that accurate reporting of findings takes precedence over recognition of success of an individual. Similarly the value of findings should be based on their intrinsic value and the credibility of the research.

Candidates will be expected to:

- understand the way in which the scientific community assesses new contributions to a body of research
- understand the process of peer review as a pre-requisite to publication
- understand the role of professional journals.

Examples in this specification include the following:

- *Students might compare a published journal article on some aspect of child development with an unsubstantiated magazine article (§3.4.3).*
- *Candidates might discuss the role of peer review in validating research (§3.1.2).*

L Appreciate the ways in which society uses science to inform decision-making.

Scientific findings and technologies enable advances to be made that have potential benefit for humans.

In practice, the scientific evidence available to decision makers may be incomplete.

Decision makers are influenced in many ways, including their prior beliefs, their vested interests, special interest groups, public opinion and the media, as well as by expert scientific evidence.

Candidates will be able to appreciate that scientific evidence should be considered as a whole. They should realise that new scientific developments inform new technology. They should realise the media and pressure groups often select parts of scientific evidence that support a particular viewpoint and that this can influence public opinion which in turn may influence decision makers. Consequently, decision makers may make socially and politically unacceptable decisions based on incomplete evidence.

Candidates will be expected to:

- understand how psychological research has contributed to decision-making in such areas as education and child-care provision, health and the law.

Examples in this specification include the following:

- *Students might use the internet to find out about*
 - (i) *psychological research into eyewitness reliability (Loftus et al) (§3.3.4);*
 - (ii) *the Devlin Report 1976 about the use of uncorroborated eyewitness accounts in criminal proceedings (§3.3.4).*
- *Candidates might discuss the effect of research into attachment on local authority and private child-care provision and changes in adoption practices (§3.3.1).*